

Residential Permitting and Construction under Environmental Litigation Risk

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Abstract

What is the optimal level of public participation in the development process? State environmental policy acts (SEPA) across the United States require some developers to publicly review the potential environmental impact of their projects. SEPA in some tight housing markets like New York and California also apply to government permitting, so local citizens can use SEPA to sue residential developers, delaying or preventing development. Lawsuits impose additional costs on developers and introduce delays and uncertainty, so a greater risk of litigation could deter developers who would have otherwise built more housing. Using building permit application data, city housing stock estimates, legislative histories, and state case law regimes, I analyse patterns of residential development under different levels of SEPA litigation risk 1) historically across states; and 2) recently in Seattle and western Washington state. I find that states with the highest counts of cases brought against residential developers under SEPA saw large historical reductions in the permitting rate for housing units in large multifamily buildings with over 5 housing units. A preferred estimate of just over one fewer unit of this type permitted per 1,000 population extrapolates to over 850,000 housing units that would have been permitted in California from 1960-1999. Other estimates, including of the effects of a recent environmental review holiday in Seattle, are not statistically different from a null effect but suggest that litigation risk is a meaningful deterrent of residential permitting and development.

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I. Introduction

What is the optimal level of public participation in the development process? Answering this question requires an understanding of the impact of public information sharing requirements on developers. For an example of a relevant policy space, recent public sentiment and legislative action suggest there is a housing supply crisis in the United States.¹ Housing under-supply in certain parts of the country has been credited with reducing national productivity by several percentage points (Hsieh & Moretti, 2019) and may have adverse climate effects (Glaeser & Gyourko, 2018). Many of the most severely supply-constrained housing markets in the United States are located in states that have passed environmental impact review (hereafter EIR) requirements through State Environmental Policy Acts (hereafter SEPAs). These laws differ from the kinds of direct land-use regulations that are studied in the urban economics literature; instead of prescribing standards that apply directly to development projects², SEPAs require developers to review the potential environmental impacts of proposed projects and share those findings with the public. Some SEPAs also expose developers to the risk of a lawsuit – based on their public EIRs – carried out by local residents who oppose the development. To what extent do residential developers respond to this litigation risk?

Understanding the relationship between litigation risk and homebuilding patterns is crucial both broadly for informing the study of optimal public participation in government and narrowly for designing effective policy to address housing shortages. To understand why, it is useful to take as a case study the strictest and most criticized SEPA, California’s CEQA. Since the California Supreme Court’s 1972 decision in *Friends of Mammoth v. Mono County*, state courts have interpreted CEQA as imposing far-reaching environmental review requirements for any government “project”, including private development for which local government *only* supplies a permit. The law also grants standing for legal action to nearly any person or group that can demonstrate either any potential harm to them or a relevant “public interest in environmental quality.” There are therefore two mechanisms through which CEQA impedes residential developers: first is a costly environmental review that is almost always required but can be consistently planned for, and second is the chance for a long, costly lawsuit that is difficult to predict. Policymakers hoping to spur building through SEPA reforms should focus on policy adjustments that target the factor (review requirement or risk of lawsuit) that is most impactful; the empirical work in this paper informs this question. Policymakers in other fields can then apply the insights provided by this analysis to other questions about optimal public input.

Residential permitting under litigation risk therefore represents an underserved area of study, especially given the size of the housing market. The residential development process involves investing significant resources up front to acquire land and permits. Outside of the housing market, finance research describes the effects of litigation risk on firm behavior: firms at greater risk of lawsuits disclose earnings earlier (Skinner, 1997), hold more cash (Oliviero et al., 2024), and issue fewer error-based financial restatements (Donelson & Yust, 2014). These decisions are consistent with firms behaving more cautiously under greater risk of a lawsuit; if the same logic holds for residential developers, they would be less likely to invest in new development under a SEPA than otherwise. However, the role of litigation risk on capital investment itself is not clear; in certain cases, greater (Arena & Julio, 2015) and lesser (Chen et al., 2020) litigation risk have both predicted declines in future investment. If land and permit procurements function only as capital investments for residential development firms, the role of litigation risk in those processes becomes less clear. Do those firms under greater SEPA litigation risk maximize the future value of the option to build embedded in permits (Cortes & LaPoint, 2026) by procuring more of them, or do they act cautiously and avoid incurring the risk associated with a marginal permit by procuring fewer of them? Do they change the types of projects they pursue? And, if residential developers do adjust permitting behavior under litigation risk, do they readjust when legislators carve out SEPA exemptions for residential development?

This paper contributes to the housing supply and litigation risk literatures by examining residential permitting and construction responses to changes in litigation risk. Three trends in SEPA legislation provide the relevant context for analysis. First, the initial wave of SEPAs in the early 1970s provides a plausibly

¹See, for example, the unusually strong bipartisan support behind the 21st Century ROAD to Housing Act, alongside the modern SEPA fixes discussed further in this paper.

²For example, minimum lot size requirements or parking minimums.

exogenous shock to builder litigation risk, as the initial motivation behind SEPA was not typically understood to be impeding private development – rather, they aimed to build transparency into the development process and were used as legal tools in the late “freeway revolts” of the 1970s. Second, since 2024, the six states with the highest SEPA case counts to date have all passed “fixes” to their SEPA designed to reduce or remove the threat of SEPA litigation for residential developers. Third, a 2023 SEPA carveout in Washington state that applied only to residential construction in Seattle provides an opportunity to compare developer responses to the removal of SEPA litigation risk without being complicated by other across-state differences in the permitting process, and there is reason to believe that developers in Seattle did not anticipate this “environmental review holiday.” The first two trends inform the historical analysis. The third trend provides grounds to check the conclusions derived from study of the historical context. For example, builders may respond more strongly to an immediate full release from litigation risk like the one caused by the Seattle carveout than they responded to the introduction of litigation risk caused by SEPA passing because the full brunt of litigation risk was developed over time in state courts.

The study of regulation and the housing supply increasingly relies on creative methods to measure the differences between similar laws in different places. Gyourko and Molloy (2015) give particular credit both to the use of creative measures of regulation (as in Brooks and Lutz (2014)) and to the helpfulness of difference-in-differences approaches in following changes in regulation over time (as in Cunningham (2007) and Kahn et al. (2010)). This is necessary for analysing SEPA, as even though California and Montana both technically have SEPA on the books, the two policies differ in important ways. I use case law regimes obtained from a popular online legal database to proxy the risk to builders inherent in each of the different SEPA, relying on court outcomes like case counts litigated under a SEPA and whether an initial SEPA suit successfully paused residential development to group states with similar levels of environmental litigation risk together. These legal outcomes combine with work by Ma et al. (2009) to provide a comprehensive classification of SEPA and other EIR related policies for all 50 states. This classification provides one of the sets of variation in litigation risk that underpins the model on which my analysis relies. The other set of variation comes from the six states that have passed state-wide “fixes” to their SEPA since 2024 designed to lessen or remove litigation risk from residential developers.

In addition to studying permitting patterns in the historical United States, I leverage a modern case study at the city level. In July 2023, the Washington state legislature passed an amendment to their SEPA that exempted residential development in Seattle from all SEPA review, thereby freeing developers from the risk of a SEPA lawsuit. Comparing estimates of the housing stock during Seattle’s review holiday to those of other cities within close proximity, with similar geographies, or under the same appeals court provides about as much isolation of the impacts of removing SEPA litigation risk from other differences in local land use controls as is possible in the United States. State-provided data estimates leverage permitting data along with construction and demolition records to calculate more accurate housing stock estimates than is possible in the historical specification, improving the relative precision of my modern estimates.

I employ a number of modern empirical strategies that mitigate concerns about parallel trends in home permitting data across states and municipalities. For my historical analysis, I employ the synthetic difference-in-differences strategy of Arkhangelsky et al. (2021) and Clarke et al. (2024) to accomodate staggered SEPA adoption and potential heterogeneity in SEPA impacts. To do so, I rely on federal residential permitting data from the Census Bureau’s Building Permits Survey. The historic data from the BPS covers state-years extending back to 1960, reaching far enough before the SEPA wave began in 1970 to evaluate pre-trends. For the Seattle analysis, official population and housing estimates provided by Washington state incorporate modern permitting, construction, and demolition data to provide a detailed picture of the housing supply in Washington state before and during the Seattle review holiday. A synthetic control analysis (Abadie, 2020; Abadie et al., 2010) provides insight into developer responses to the policy.

I find that the six states that have passed modern statewide SEPA fixes experienced a decrease in units permitted in large multifamily developments by a preferred estimate of -1.073 units per 1,000 population per year between the passage of their SEPA and the year 1999. This is an enormous result. For example, extrapolated to the counterfactual, this estimate suggests that California would have permitted 850,892 more residential units in large housing structures (e.g. condos, apartment complexes) over that time span than they actually did. Many estimates of the current housing shortage in California fall below this

number (including a 2020 Freddie Mac estimate of 820,000 missing units) (Christopher, 2025), suggesting that SEPA litigation risk was a significant deterrent to residential permitting in California. The other five states in the “fixers” group would also have seen a significant bump to their housing supply under the same counterfactual. In states that have had SEPA cases brought against residential developers, estimates of housing units permitted in all kinds of structures (including single family homes) are uniformly negative, albeit statistically underpowered. In SEPA states with no history of SEPA being used to sue residential developers, these estimates are often positive and broadly follow development trends in non-SEPA states.

The synthetic Seattle is also statistically underpowered across all three potential sets of control cities. Nevertheless, the treatment effect estimates themselves are uniformly positive and quite large: the median synthetic control estimate of the effect of Seattle’s review holiday is an average increase in 8.957 housing units per 1,000 people in Seattle per year for 2024–2026. Extrapolating to the statistically unlikely counterfactual where this estimate is indeed the true effect suggests that the Seattle review holiday added over 21,800 housing units to the Seattle housing supply in just three years, or about half of the average growth of the housing supply between 2020–2023 and 2024–2026. A synthetic difference-in-differences analysis of the same review holiday returns a more conservative median estimate of 5.669 housing units per 1,000 people, or about 13,800 housing units in a similarly underpowered counterfactual analysis.

The results presented in this paper constitute evidence that litigation risk from SEPAs is a powerful deterring force in the residential construction sector. Likely due to increased uncertainty in total costs, development timelines, and project feasibility, developers respond to the threat of a SEPA lawsuit like firms in other fields respond to similar legal threats: they take a cautious approach by investing less, permitting less, and building less. Policymakers interested in spurring housing development in SEPA states could therefore consider several policy options. They may consider implementing an exemption for residential construction as exemplified by Seattle’s review holiday or New York’s 2026 carveout for previously disturbed sites. If they value preserving the public reporting requirements that are integral to the intent of the SEPAs, they may consider the more measured approach taken in Minnesota in 2025 when lawmakers reduced the set of people with legal standing for a SEPA lawsuit from any resident of the state to residents of the same or bordering counties in which a project is taking place. They may even simply impose short statutory time limits for SEPA challenges, as Connecticut did in 2025. In any case, these results strongly suggest that any policy effort that reasonably reduces uncertainty in development cost or timelines would be reasonably expected to increase developers’ willingness to develop residential housing in SEPA states.

The full paper proceeds as follows. Section II models the permitting and development process, reviews the history of environmental impact review requirements in the United States, and provides context relevant to the Seattle review holiday. Section III discusses the data and provides summary statistics. Section IV explains the identification strategies for each set of policy changes I study. Section V presents the results of estimating each model and corresponding robustness checks, and section VI concludes.

II. Model and Background

II.1 Modeling the Developer’s Decision

Residential development businesses should be expected to internalize the regulatory environment under which they plan to operate (Khan et al., 2024; Luger & Temkin, 2000). SEPAs that open developers to litigation risk for their environmental reviews introduce uncertainty to the total costs and timeline associated with the decision to develop. Mayer and Somerville (2000a) show that regulations that delay a project or make its viability uncertain are the most prohibitive of efficient construction supply. Indeed, Gyourko and Molloy (2015) cite the early suspicions of SEPAs expressed by Frieden (1979) as they describe the move from the relatively unregulated development environment prior to the 1970s (Gyourko et al., 2013) to the high-regulation development environment in the United States today. Recent evidence from another industry also shows that intellectual property firms at lower risk of legal action are more productive (Bereskin et al., 2026). Together, the uncertainty and productivity findings suggest that the risk of a lawsuit created by

SEPAs may have played a role in some states’ housing crunches by discouraging residential developers. The main purpose of this paper is to investigate to what extent this is true.

I take as my base case a representative residential development firm. The firm maximizes expected profits and takes housing market demand, the local regulatory environment, and their available resources as given. Based on these resources and constraints, the developer is assumed to understand the relative costs of land, materials, and labor at which the decision to develop new residential construction would be profitable. In this model, I abstract away from specifically modeling uncertainty in future market demand, development timelines, legal challenges, and so on. Instead, entrenched in the decision to develop is an aversion to uncertainty. Figure 1 adapts figures from previous work detailing the practical process of residential development (Luger & Temkin, 2000; Mayer & Somerville, 2000a). In both, the authors conclude that delay and uncertainty in developer timelines discourage development. Highlighted by exclamation points in Figure 1 are the two steps in the development timeline in which uncertainty and delay associated with the risk of an environmental lawsuit enter the decision in this model, as they are the steps at which developers may encounter EIR requirements.³

In the context of this paper, a SEPA that opens the firm to substantial litigation risk would be factored into the initial decision to develop only if the SEPA goes “deep enough”; the developer’s project must be subject to EIR requirements, and the SEPA must give standing for a lawsuit to interested opponents. Litigation risk enters at the stages of the development process covered by the SEPA – in this case, the two permitting stages where EIR determination is typically required. In the event of litigation at either of the two permit stages, a developer would be subject to both increased costs (spent on lawyers, supplemental EIR, etc.) and to a delayed development timeline. Litigation therefore adds both delay and uncertainty, the exact type of challenge that Mayer and Somerville (2000a) suggest is most burdensome on developers. Developers are assumed to fully internalize these risks without error, viewing an increase of litigation risk as weakly decreasing the expected profits of choosing to develop. The key question is whether this decrease in the expected profit of a given project drives developers to diversify their project portfolio by trying to secure more permits or causes them to retreat from projects and procure fewer permits.

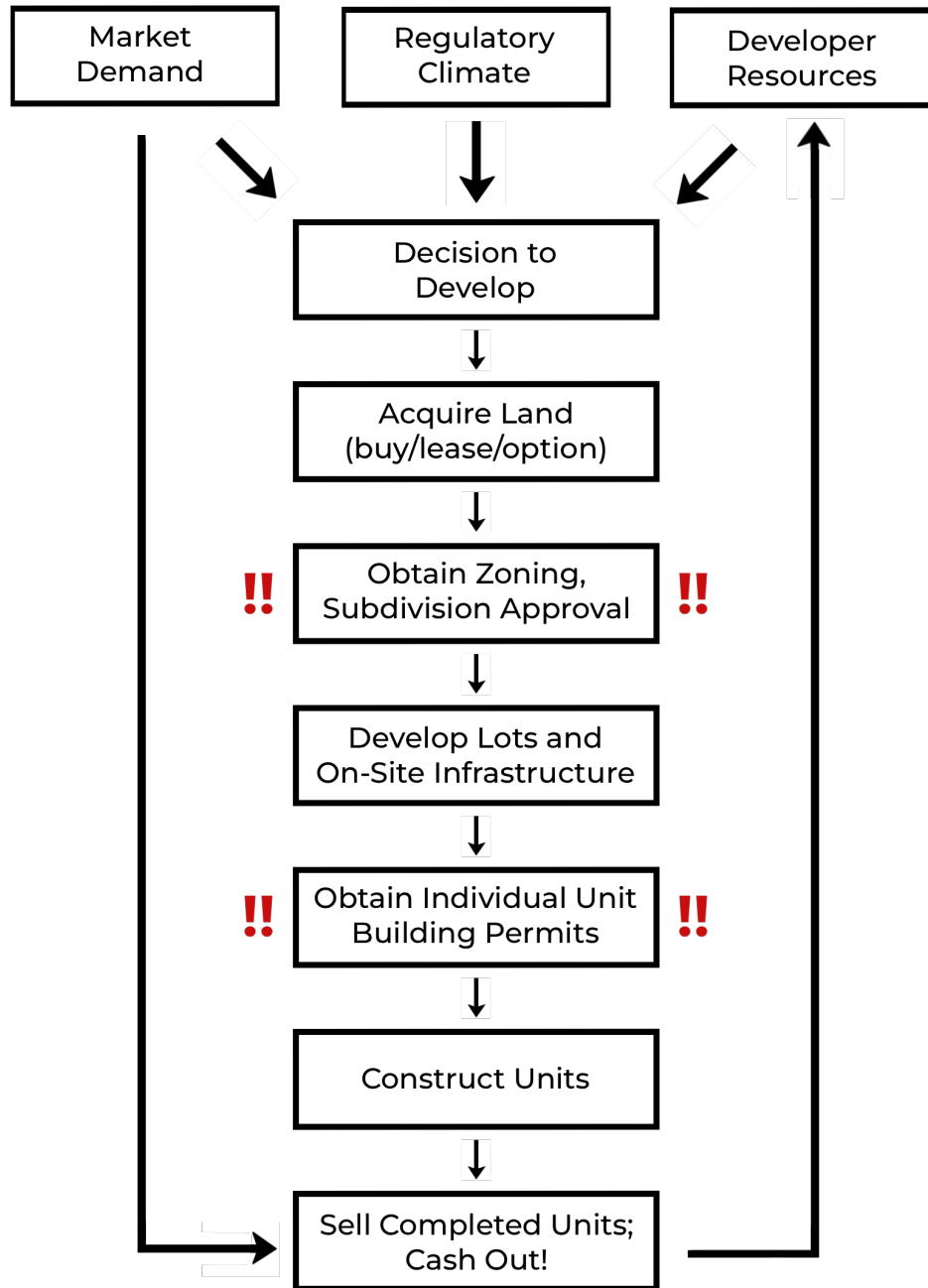
I simplify the analysis in this paper by not modeling the decision of local community members to oppose residential development for two reasons. First, to do so would require capturing the relative effectiveness of “second-best” opposition tactics within and across every state. This is prohibitively difficult given the substantial heterogeneity in local land control laws present throughout the United States, even within individual states. Second, previous research gives reason to believe this modeling is not necessary. Mayer and Somerville (2000b) use an urban growth model to show that “new housing construction [is] a function of *changes* in house prices and costs rather than ... the levels of those variables” (emphasis added). That model of the developer’s decision relies on the assumption that developers can perfectly internalize the regulatory environment under which they build; this would include the full set of legal tactics available to their opponents and the risk to the developer inherent in each option. Therefore, I assume that non-SEPA changes to the regulatory environment that would affect locals’ ability to oppose residential development are as good as randomly assigned across states and throughout years. With the problem made more tractable, I can investigate the effects of a change in litigation risk as a simple change in expected total cost of development caused only by the passage of a single type of policy.

II.2 Passing and “Fixing” the SEPAs

The National Environmental Policy Act (NEPA) went into effect January 1, 1970 in the midst of a wave of environmental activism spurred by the Santa Barbara oil spill, the Cuyahoga River fire, and other environmental disasters. The law innovated environmental impact reviews (EIR) as a tool to encourage environmental responsibility in federal decisionmaking. NEPA “requires federal agencies to assess the environmental effects of their proposed actions” by preparing reports for projects with the potential to significantly affect the environment, where “proposed actions” included permitting, construction, and land management (Agency, 2025). The law also required that these reports be shared with the public to solicit their input.

³Land prices and worker availability are assumed to be predictable for the firm. In this model, that translates to no uncertainty in the Acquire Land, Develop Lots, and Construct Units steps.

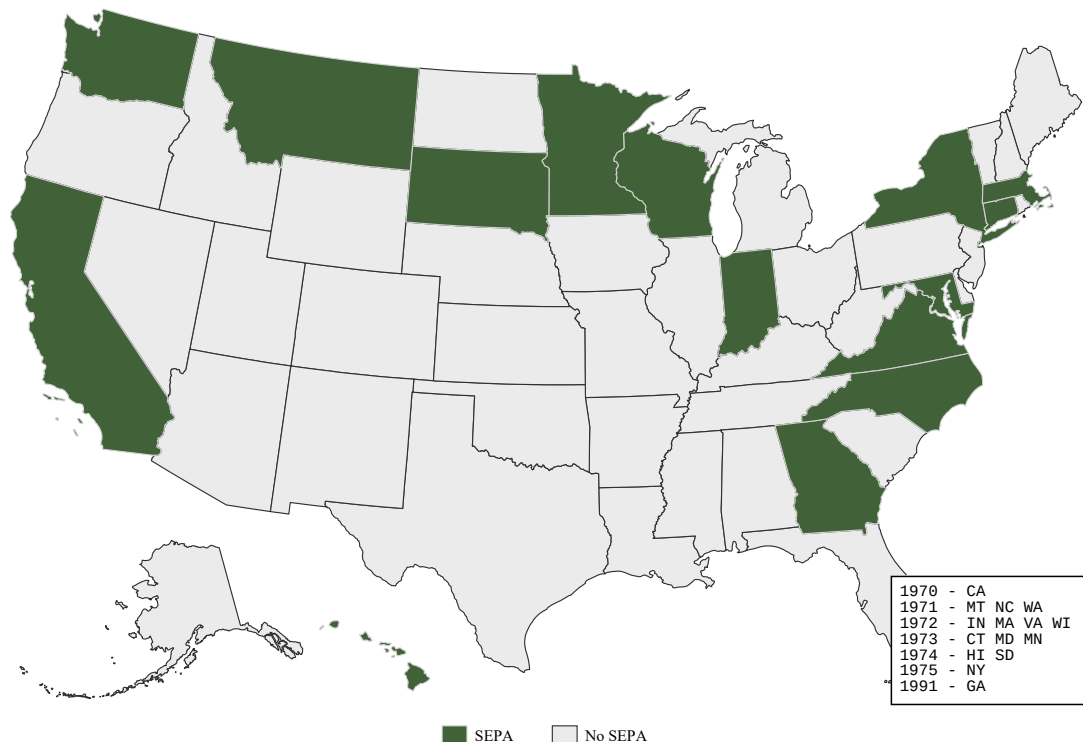
Figure 1: The Developer's Decision Calculus



Notes. This figure adapts and combines Figure 1 from Mayer and Somerville (2000a) and Figure 2 from Luger and Temkin (2000). Exclamation points denote stages of the process in which legal challenges are most likely to arise.

NEPA was soon followed by the passage of state environmental policy acts (SEPA) in several states, as shown in Figure 2. Though many of the SEPA passed in this era closely resemble NEPA, there are key differences across states in their scope and regulatory burden.

Figure 2: States with SEPA



Notes. Note box lists states whose SEPA went into effect in a given year. Executive orders that impose EIR requirements are not treated as SEPA due to a lack of case law.

The two most relevant aspects of the SEPA to private developers in a given state are 1) whether a SEPA applies to their project at all; and 2) how likely they are to be taken to court under it. Existing legal and environmental planning research offers insights that will enable the kind of creative measures of regulation encouraged by Gyourko and Molloy (2015). In a summary of the evolution of case law around SEPA, Selmi (2006) explains that “a lawsuit challenging a project approval has a very real effect” even if the lawsuit is unsuccessful because “banks are cautious about financing a project that lies under a legal cloud.” The paper also notes that such an effect is “arguably inefficient, for most SEPA litigation is ultimately unsuccessful” in obtaining legal relief. The model of developer decision-making I rely on in this paper suggests that developers are able to accurately internalize the litigation risk present in the regulatory environment under which they intend to work. Therefore, it is worthwhile to pursue a measure of litigation risk associated with SEPA by examining each SEPA’s early case law outcomes. To that end, Table 1 lists every state with a SEPA, the section of that SEPA cited most frequently in court decisions reported in West’s National Reporter System (NRS), the number of court decisions in the NRS that cite that section of the SEPA, and (if applicable) the reference, year, and outcome of the earliest case in the NRS decided under that SEPA that concerned plans for residential development.

Table 1 makes it clear that there are three “blocks” of historical SEPA litigation risk: first, the four states whose earliest residential SEPA case succeeded in blocking development (California, New York, Washington, and Minnesota); second, the four states whose earliest residential SEPA case failed (Massachusetts, Connecticut, Wisconsin, and Hawai’i); and third, the seven states whose SEPA have never been used to bring legal action against a planned residential development (Montana, North Carolina, Maryland, South Dakota, Georgia, Indiana, and Virginia). It is reassuring that this first block aligns exactly with those iden-

Table 1: States, SEPA, and Early SEPA Cases

State	Authority	Cases	Earliest Residential Case	Year	Success?
California	Cal. Pub. Res. Code § 21000	1,874	Friends of Mammoth v. Mono County	1972	Yes
New York	N.Y. Env'tl. Conserv. § 8-0101	928	Tuxedo Cons. & Taxpayers v. Tuxedo	1978	Yes
Washington	Wash. Rev. Code Ann. § 41.21C.030	193	Eastlake Council v. Roanoke Assoc.	1973	Yes
Minnesota	Minn. Stat. Ann. § 116D.04	148	Cedar-Riverside Env. Def. Fund v. Hills	1976	Yes
Massachusetts	Mass. Gen. Laws Ann. Ch. 30 § 61	107	Springfield Y Trust v. Exec. Dir. MHFA	1976	No
Connecticut	Conn. Gen. Stat. Ann. § 22a-1	76	Red Hill Co. v. Glastonbury Cons. Com'n	1989	No
Wisconsin	Wis. Stat. Ann. § 1.11	64	Wis. Env. Decade v. ILHR Dept.	1981	No
Hawai'i	Haw. Rev. Stat. § 343-5	63	McGlone v. Inaba	1981	No
Montana	Mont. Code Ann. § 75-1-201	29	N/A	N/A	N/A
North Carolina	N.C. Gen. Stat. § 113A-4	19	N/A	N/A	N/A
Maryland	Md. Nat. Res. § 1-301	11	N/A	N/A	N/A
South Dakota	S.D. Codified Laws § 34A-9-4	3	N/A	N/A	N/A
Georgia	Ga. Code Ann. § 12-16-4	2	N/A	N/A	N/A
Indiana	Ind. Code § 13-12-4-5	2	N/A	N/A	N/A
Virginia	Va. Code Ann. § 10.1-1188	1	N/A	N/A	N/A

Notes. Authority refers to the section of a SEPA cited the most times by cases reported in WestLaw as of 8 June 2026. Cases refers to the total number of reported decisions recorded in West's National Reporter System that cite the listed Authority. First Residential Case refers to the earliest case reported in the NRS that both 1) cited the given Authority and 2) involved plans to develop new residential units, where Year reports the year of the decision in that case and Success? denotes whether the final decision reversed issuance of a permit or otherwise further impeded the planned development.

tified in Figure 1 of Pendall (1998) as having a “local [EIR] mandate”, the only kind that would apply to most residential developers. The legal scholarship on SEPA broadly concludes that these “local mandate” states impose a significant burden on developers by not limiting their EIR requirements under their SEPA to projects primarily carried out by the state (Selmi, 2006; Watts, 1995). Instead, by requiring EIR of projects whose only interaction with government is permit acquisition, these states lengthen the development planning process and open private residential developers to the risk of costly legal action. While developers in the other SEPA states likely need to consider those SEPA in their decision to develop, it seems unlikely that they would play as large a role in that decision as the local mandate SEPA do for developers in local mandate states. One purpose of this paper is to quantify whether and to what extent this extra consideration affected historic development permitting patterns.

Recent legislative action from several SEPA states presents another natural way to categorize states’ SEPA litigation risk. Since 2024, six states with SEPA passed legislation to reduce or eliminate the uncertainty associated with the risk of an EIR lawsuit for residential developers. Table 2 lists the relevant states, laws or regulatory codes, dates, and types of development associated with these “fixes.”

Table 2: Recent SEPA “Fixes”

State	Updated Code	Effective Date	Note
Minnesota	Minn. Stat. Ann. § 116D.04	1 July 2025	Nearby county residence requirement
Washington	Wash. Rev. Code Ann. § 41.21C.229	27 July 2025	Transit-oriented development exempt
Connecticut	Conn. Gen. Stat. Ann. § 22a-19	1 October 2025	30 day max for residential hearings
California	Cal. Pub. Res. Code § 21080.66	11 October 2025	Urban density exempt
Massachusetts	301 Code of Mass. Regs. 11.02	30 January 2026	Any dense housing exempt
New York	N.Y. Env'tl. Conserv. §§ 8-0105.11 & 8-0111.5-a	26 May 2026	Previously disturbed sites exempt

Notes. Updated Code refers to the regulation or law that was updated. “Previously disturbed sites” is nearly synonymous with urban infill.

The end goal of passing these SEPA fixes is clearly to spur residential development in high-demand areas. Though it seems obvious that mitigating or removing the risk of a SEPA lawsuit for private developers

would lead them to develop further, these fixes may not have the intended effect if developers do not believe they will last or if local development opponents find loopholes in the new policies. It is reassuring that the six states that passed these broad SEPA fixes are the six states with the highest recorded number of reported SEPA lawsuits, suggesting that the types and judicial interpretations of SEPAs in these states are perceived as more onerous for developers generally than those in other states. These updates are themselves too recent to be analyzed with available data, but this will change as time goes on. For now, their use as a further SEPA classification tool is sufficient.

II.3 Seattle’s Environmental Review Holiday

A unique SEPA carveout not listed in Table 2 was passed in 2023 in Washington state, and it is worth its own separate investigation. On 23 July 2023, Washington’s SEPA was updated by the state legislature to read as follows:

Until September 30, 2025, all project actions that propose to develop one or more residential housing or middle housing units within a city west of the crest of the Cascade mountains with a population of 700,000 or more are categorically exempt from the requirements of this chapter.

Wash. Rev. Code Ann. § 41.21C.229(4)

Seattle is the only city in Washington that meets these criteria. This single-city carveout – referred to in one source as an “environmental review holiday” (Trumm, 2026) – was particularly lenient, fully exempting practically all residential development within the city boundaries. It affords a uniquely powerful opportunity to compare the effects of changes to SEPA litigation risk in as close to a unified regulatory environment as possible. In contrast with the fixes outlined in Table 2, this SEPA change is not too recent to analyse, with three years of data having already been made available since its passage. Also, there is some evidence that permitting patterns were not changing in anticipation of the holiday: projects that were under SEPA review at the time were allowed to withdraw the SEPA component of their applications (SDCI Community Engagement, 2023), suggesting that at least some developers were not anticipating this holiday when making the decision to begin the development process.

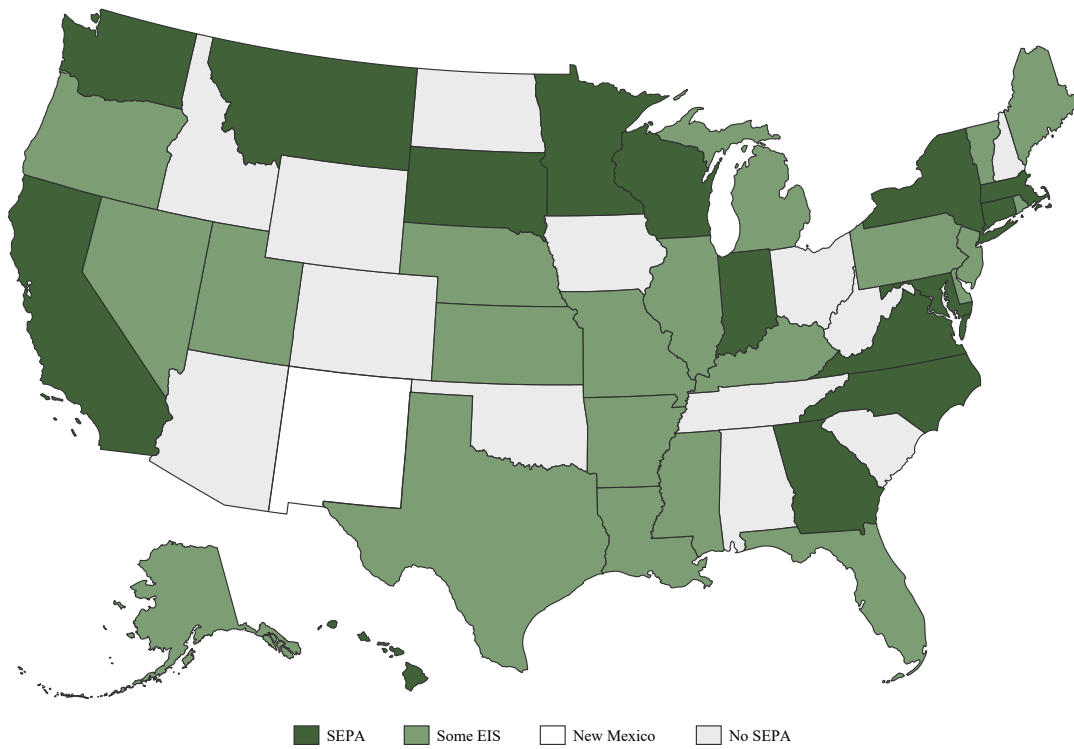
III. Data and Summary Statistics

Building permits are public records in the United States. Since 1959, the Census Bureau has tracked building permits issued across the United States using the Building Permits Survey, a monthly survey conducted of jurisdictions within the United States that issue building permits. The historical data available to download from the Census Bureau track state permit totals and extend from 1960 to present day. The sampling frame for the survey covers all “permit-issuing places” that do not double count. For example, if a city extends zoning permits and a county extends building permits for that city, only the county permitting office is surveyed. According to the Census Bureau, more than 99% of all privately-owned residential buildings are constructed in permit-issuing places.⁴

Of the blocks of SEPA case law described in Table 1, it is not obvious *ex ante* which block(s) should constitute “treatment” for developers in this study. It would also be an error to avoid addressing the limited EIR requirements some states impose on builders in sensitive areas like coastal plains. There is also the question of which states will eventually serve as the most appropriate comparison group to the SEPA states. Should states like Virginia that pass an underpowered SEPA be included in the same treatment group as California, the strictest SEPA state? Should states with limited EIR requirements like Nevada be included in the same pool of control states as Wyoming, where there are no EIR requirements at all? Figure 3 displays the SEPA states alongside the states identified by Ma et al. (2009) as having limited and no EIR requirements, with a few adjustments described in later sections.

⁴A detailed explanation of the methodology behind the Building Permits Survey, as well as notes about imputation, reporting, and other data quirks associated with the BPS permitting data, can be found at U.S. Census Bureau (2026).

Figure 3: SEPA and EIR Requirement States



Notes. Executive orders that impose EIR requirements are not treated as SEPAs due to a lack of case law. New Mexico passed a SEPA in 1971 but repealed it in 1974; it now has limited EIR requirements but is excluded from my historical analysis.

I adjust the classification of two states from their presentation in Ma et al. (2009). First, while they treat New Jersey as having a SEPA due to a state executive order, I always treat New Jersey as having a limited EIR requirement. An executive order cannot give a plaintiff standing under which to bring a legal case against anyone but those carrying out the executive order, so finding case law for New Jersey’s “SEPA” is not possible, which warrants reclassification.⁵ Second, I exclude New Mexico from all historical specifications. As explained in McCash (1974), New Mexico passed a SEPA statute in 1971 which it fully repealed in 1974. Turning treatment on and back off again is not compatible with the identification strategy I plan to use in this historical specification, so New Mexico must be excluded.

Table 3 displays summary statistics over relevant permit outcomes for the three blocks of SEPA states and the two blocks of non-SEPA states shown in Figure 3, broken down across pre-SEPA and post-SEPA time periods. It shows that, adjusted for population, states with SEPAs permitted housing at similar rates as their non-SEPA peer states. However, states whose SEPAs had residential case law (blocks 1 and 2) were more inclined to build multifamily housing than other states. The statistics reported in Table 3 do not show whether the litigation risk associated with the passage of the block 1 and 2 SEPAs changed these outcome variables, but the averages do suggest that SEPAs in these states could have been intentionally written by legislators or interpreted by judges with the goal of bringing permitting trends in line with the rest of the country. Whether this heightening of litigation risk affected these permitting patterns or not is the subject of the historical section of this paper.

To accommodate the Seattle holiday specification, data on the housing supply and population of cities in western Washington is necessary. Washington state law assigns the responsibility of determining the populations of cities and towns within the state every year on April 1 to the state Office of Financial Management (Washington State Office of Financial Management, 2026). These population estimates are accompanied by estimates of the housing stock obtained using housing construction, completion, and demolition data reported by cities and counties to the state office.⁶ Using estimates that incorporate completion and demolition data allows me to estimate the effect of SEPA changes on the housing stock as a whole, instead of being limited to just permitting as in the historical data. However, before 2020, official city-level estimates are not broken down by the type of housing unit, so the Seattle analysis is limited only to total housing unit counts. Despite this limitation, the extra information from construction and demolition data allow for richer examination of the effects of environmental litigation risk on the overall housing stock than would be possible with just the historical permitting data.

Comparing the housing stock during Seattle’s environmental review holiday to that of other incorporated municipalities within similar locales provides about as clean a test of the impacts of removing SEPA litigation risk as is possible in the United States, as it avoids many of the pronounced regional differences in land use regulations that are omnipresent across the many states of the nation. Nevertheless, the question of what cities constitute an appropriate comparison group is not easy to answer. Would the most appropriate comparison group be regional cities in the three counties that make up the Seattle-Tacoma-Bellevue metropolitan area? Perhaps the restrictive forces of building in sensitive coastal environments is most at play, so we should consider the set of cities in the fifteen counties in western Washington that border either the Puget Sound or the Pacific Ocean? Perhaps, owing to the fact that judicial interpretation of SEPAs plays a large role in the regulatory environment faced by builders, we should consider the six counties that make up Division I of the Washington Court of Appeals, where a cities’ SEPA cases would meet the same panel of judges as a Seattle SEPA case would?

Table 4 displays summary statistics of relevant outcomes for Seattle and the non-Seattle cities in three potential donor pools: the Seattle metro area, Washington counties that border the Pacific Ocean or the Puget Sound, and the counties in Division I of the Washington Court of Appeals. The data in this table make it clear that Seattle is the largest city in this region by a good margin, but the averages displayed overstate this comparison. Included with equal weight in the reported averages are very small towns with only a few thousand residents each; many more populous cities like Tacoma and Bellevue build at rates

⁵Michigan, Texas, and Utah all also passed SEPA-like executive orders which have since been repealed (Pendall, 1998), but they are already classified as Tier 2 states due to their remaining limited EIR requirements.

⁶The OFM city population and housing stock estimates are available through the State of Washington Open Data site at data.wa.gov as the (Housing Units/Population) by State, County and City, 1990 to Present datasets. Accessible as of 27 July 2026, with data also available in the replication package for this paper.

Table 3: Historical Building Permit Summary Statistics by Block

	Total Permits	Permits/1K	SFH/1K	Mid/1K	Big/1K
<u><i>SEPA Fixers</i> (CA CT MA MN NY WA)</u>					
Average, 1960–1999:	74,021.31	6.52	3.91	0.51	2.10
1960–1975	81,108.94	7.41	4.04	0.62	2.76
1976–1999	69,296.23	5.92	3.83	0.43	1.66
<u><i>SEPA Block 1</i> (CA MN NY WA)</u>					
Average, 1960–1999:	56,360.77	5.98	3.65	0.43	1.90
1960–1975	62,728.08	7.05	3.86	0.53	2.66
1976–1999	52,115.89	5.27	3.52	0.36	1.40
<u><i>SEPA Block 2</i> (CT HI MA WI)</u>					
Average, 1960–1999:	19,256.62	6.33	3.68	0.38	2.27
1960–1975	21,942.08	8.24	4.20	0.47	3.58
1976–1999	17,466.31	5.05	3.33	0.32	1.40
<u><i>SEPA Block 3</i> (GA IN MD MT NC SD VA)</u>					
Average, 1960–1999:	28,971.29	6.28	4.26	0.29	1.73
1960–1975	24,815.12	6.23	3.72	0.26	2.25
1976–1999	31,742.07	6.32	4.63	0.32	1.38
<u><i>Limited EIS Block</i> (21 states)</u>					
Average, 1960–1999:	26,985.69	6.27	4.00	0.46	1.81
1960–1975	25,186.83	6.53	3.85	0.49	2.19
1976–1999	28,184.93	6.10	4.10	0.45	1.55
<u><i>No EIS Block</i> (13 states)</u>					
Average, 1960–1999:	17,207.62	5.99	4.01	0.40	1.58
1960–1975	15,575.29	5.92	3.77	0.42	1.73
1976–1999	18,295.83	6.03	4.16	0.39	1.48

Notes. Base statistics are permits per year in a given state. Transformed permit outcomes are per capita based on population estimates in thousands (/1K = per 1,000) and averaged over states in a given block across the given time range. Blocks are as listed or following Figure 3. Mid = permitted units in buildings with 2-4 units. Big = permitted units in buildings with 5 or more dwelling units. Yearly population estimates come from Census Bureau estimates as reported by FRED; California estimates from FRED series CAPOP, Alabama from ALPOP, and so on.

similar to Seattle. The mix of geographically similar smaller towns with higher per 1,000 housing measures and other relatively large cities with lower per 1,000 statistics lends itself well to the identification strategy described in the next section.

IV. Identification Strategy

IV.1 Synthetic Difference-in-Differences

A standard difference-in-differences approach is the default starting point for evaluating the effects of changes in state policies over time. This approach relies on the parallel trends assumption that outcomes in treated states 1) followed trends in untreated states before treatment; and 2) would have continued to follow the same trend as outcomes in untreated states had they been left untreated. This assumption is unlikely to hold in this context; there were likely many changes in housing trends across disparate regions of the United States, even if large ones like the spread of residential air conditioning had already promulgated by the

Table 4: Washington State Housing Supply Summary Statistics by Comparison Group

	Population	Total Units	Units/1K
<u><i>Seattle</i></u>			
Average, 2010–2026:	707,476.00	358,536.18	506.24
2010–2019	656,827.70	328,728.90	500.79
2020–2023	755,278.75	383,535.50	507.70
2024–2026	812,566.67	424,561.33	522.45
<u><i>Seattle-Tacoma-Bellevue MSA</i></u>			
Average, 2010–2026:	26,487.55	10,889.50	412.42
2010–2019	25,102.95	10,381.53	414.68
2020–2023	28,043.95	11,378.32	407.68
2024–2026	29,027.65	11,930.95	411.24
<u><i>Coast or Sound Counties</i></u>			
Average, 2010–2026:	19,893.85	8,356.89	438.50
2010–2019	18,838.22	7,958.53	441.78
2020–2023	21,055.88	8,733.39	432.10
2024–2026	21,863.21	9,182.75	436.13
<u><i>Appellate Division I</i></u>			
Average, 2010–2026:	23,807.53	9,813.28	425.67
2010–2019	22,512.15	9,330.83	427.61
2020–2023	25,243.19	10,279.10	421.42
2024–2026	26,211.24	10,800.38	424.86

Notes. Base statistics are population or housing units present on April 1 of a given year in a given city. Rows after the “Seattle” row exclude Seattle. Units/1K is per capita based on population estimates in thousands (/1K = per 1,000) and averaged over cities in a given set of counties across the given time range. Seattle-Tacoma-Bellevue MSA includes King, Pierce, and Snohomish counties. Coast or Sound Counties border either the Pacific Ocean or the Puget Sound, and the category includes Clallam, Grays Harbor, Island, Jefferson, King, Kitsap, Mason, Pacific, Pierce, San Juan, Skagit, Snohomish, Thurston, Wahkiakum, and Whatcom counties. Appellate Division I includes Island, King, San Juan, Skagit, Snohomish, and Whatcom counties.

1970s (Lester, 2015). A large modern applied econometrics literature has also pointed out the bias and interpretability issues of the standard two-way fixed effects (TWFE) estimator commonly applied to these kinds of research designs when treatment rollout is not simultaneous, as in this case (see Figure 2).

Fortunately, there is an estimator that is robust to biases from heterogeneous treatment effects, staggered treatment implementation, and imperfect parallel trends: the synthetic difference-in-differences estimator (SDID) (Arkhangelsky et al., 2021). What follows is a description of the canonical and staggered treatment adoption SDID designs drawing heavily from sections 2.1–2.3 of Clarke et al. (2024), whose accompanying Stata package I use to conduct my analysis. SDID estimates the average effect $\hat{\tau}^{\text{SDID}}$ of a binary treatment W_{it} received by unit i at time t even in the face of potentially violated parallel pre-trends. The canonical SDID estimation solves the following minimization problem:

$$(\hat{\tau}^{\text{SDID}}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \underset{\tau, \mu, \alpha, \beta}{\operatorname{argmin}} \left\{ \sum_{i=1}^N \sum_{t=1}^T (Y_{it} - \mu - \alpha_i - \beta_t - W_{it}\tau)^2 \hat{\omega}_i^{\text{SDID}} \hat{\lambda}_t^{\text{SDID}} \right\} \quad (1)$$

where the right hand side is the ATT from a standard TWFE regression with unit fixed effects α_i and time period fixed effects β_t augmented with optimally selected unit weights $\hat{\omega}_i^{\text{SDID}}$ and time weights $\hat{\lambda}_t^{\text{SDID}}$. Details on the construction of $\hat{\omega}_i^{\text{SDID}}$ and $\hat{\lambda}_t^{\text{SDID}}$ are contained in both Arkhangelsky et al. (2021) and Clarke et al. (2024); they weight units and time periods such that the estimator matches treated and untreated units

across time in a way that “synthesizes” parallel trends. In this context, μ is a constant, α_i and β_t represent state and year fixed effects, $\hat{\omega}_i^{\text{SDID}}$ weights states, $\hat{\lambda}_t^{\text{SDID}}$ weights years, and W_{it} is a binary indicator for having passed a SEPA.

As described in section III, it is not clear which “block” of SEPAs constitute “treatment.” I estimate $\hat{\tau}^{\text{SDID}}$ separately for each block of SEPA states: the four states whose earliest residential SEPA case successfully enjoined development, the four states whose earliest residential SEPA case failed, and the seven states that have never had a residential case brought under their SEPAs. I also include as a treatment group the six states that have passed modern fixes to their SEPAs. It is also true that the choice of “donor pool” control states is not obvious *ex ante*. To this end, I alternate including in the control group only those states with a limited-scope EIR requirement (Some EIR in Figure 3), only those states with no EIR requirements, and the full set of non-SEPA states (excluding New Mexico, as described in III), giving a total of twelve estimates of $\hat{\tau}^{\text{SDID}}$ for each outcome in my main historical specification.

SDID is a particularly robust estimator. It inherits consistency from canonical difference-in-differences designs when the parallel trends assumption holds. In cases where that assumption fails for unadjusted data, it finds consistency by synthesizing unit and time weights to approximate parallel trends. This relies only on a donor convex hull assumption – that the donor pool can be weighted to create a sufficiently well-fit counterfactual trend – that it inherits from its roots in synthetic control methods. Achieving consistency under just one of the two assumptions has led some to refer to SDID as “robust in a ‘double’ sense” (Nguyen, 2025), and it helps make estimated effects believable even when treatment is assigned as a function of unobserved unit characteristics.

As the outcome of interest Y_{it} , I use the number of housing units permitted per 1,000 people in a given state and year. I estimate the model separately for total units, single-family homes, units in mid-sized buildings of 2-4 units, and units in big buildings of 5 or more units. Accounting for population in the permit count outcomes mitigates the influence of baseline population differences and growth trends, which are among the largest contributors to demand in the housing market.⁷ I choose to end the time period of interest for this specification in 1999. This time period is selected to avoid potential complications from 1) other U.S. environmentalism pushes like the Regional Greenhouse Gas Initiative and the NO_x Budget Trading Program; and 2) the run-up to the housing crisis at the beginning of the Great Recession.⁸

IV.2 Synthetic Control

The classic use case for a synthetic control estimator is to estimate the effects of a treatment on a single treated unit among many, such as one region of Spain (Abadie & Gardeazabal, 2003), one state in the United States (Abadie et al., 2010), or one city in Washington state. As summarized by Abadie (2020), the synthetic control method is useful because, when comparing a single unit affected by treatment to potentially many others, “a combination of unaffected units often provides a more appropriate comparison than any single unaffected unit alone.”

For a set of cities C in western Washington state within a given time period t (here, a year), we observe an outcome of interest $Y_{c,t}$ (here, total housing units). Call $Y_{c,t}^N$ the total housing units in city c in year t when the treatment is Not active, and $Y_{c,t}^T$ the potential outcome when Treatment is active. The objective is to reliably estimate the following period-specific effect:

$$\tau_{c,t} = Y_{c,t}^T - Y_{c,t}^N \quad (2)$$

For every city but Seattle, $Y_{c,t}^T$ is assumed to equal $Y_{c,t}^N$, so that $\tau_{c,t} = 0$.⁹ For Seattle, $Y_{Seattle,t}^N$ is not observed in any time periods after treatment begins, so estimating $\hat{Y}_{Seattle,t}^N$ represents the key challenge addressed by a synthetic control.

⁷A supplemental specification will estimate the $\hat{\tau}^{\text{SDID}}$ of SEPA passage on a housing price index constructed for American cities by Lyons et al. (2026).

⁸Pending additional robustness checks around a range of potential end dates [HOPEFULLY!!] will not meaningfully affect the results.

⁹This assumes that the Seattle review holiday had no effect on construction patterns in surrounding cities.

A synthetic control Seattle is a weighted average of the other included cities’ total per capita housing counts. For a vector of non-Seattle city weights $\mathbf{W} = (w_1, w_2, \dots, w_{C-1})$ that are non-negative, sparse, and sum to one, the synthetic control estimators of $Y_{Seattle,t}^N$ and $\tau_{Seattle,t}$ are:

$$\hat{Y}_{Seattle,t}^N = \sum_{c \neq Seattle} w_c Y_{c,t} \quad (3)$$

and:

$$\hat{\tau}_{Seattle,t} = Y_{Seattle,t}^T - \hat{Y}_{Seattle,t}^N \quad (4)$$

The key problem then is how to calculate $\mathbf{W}$, the vector of city weights that build the synthetic Seattle. Several methods exist for fitting these weights, including minimizing mean squared prediction error in the pre-treatment periods (as in Abadie and Gardeazabal (2003) and Abadie et al. (2010)) and ridge penalty selection (as implemented in Clarke et al. (2024)). Regardless of the fitting method, synthetic control methods rely on a convex hull assumption to identify causal treatment effects. The assumption is that a sufficiently well-fitting control “unit” can be constructed as a convex combination of units in the “donor pool” – the set of untreated units included as potential contributors to the synthetic control. This assumption does not have a formal test, but the traditional plots and figures associated with the exposition of a synthetic control serve as a valuable check. The Seattle outcomes I study are not extreme values in my dataset, so they plausibly fall within the convex hull of other Washington cities.

It is also helpful for the credibility of a synthetic control estimate to use in the donor pool those units that would already be considered similar to the treated unit. As categorized in Table 4, I report the results of estimation using three potential donor pools (in order from fewest to most counties covered): first, the set of cities in the three counties that make up the Seattle-Tacoma-Bellevue metropolitan statistical area; second, the set of cities in the six counties in Division I of the Washington Court of Appeals; and third, the set of cities in the fifteen counties that border either the Puget Sound or the Pacific Ocean. The list of counties covered by each donor pool is reported in the notes under Table 4.

Synthetic difference-in-differences is not only useful for improving difference-in-differences estimation with synthetic control tools. It can also improve synthetic control estimation with difference-in-differences tools. Though it may be the case that a given donor pool cannot convincingly synthesize a control for Seattle by synthetic control estimation, SDID may still be able to synthesize a control with parallel trends to Seattle. To check the robustness of each synthetic control estimate, I compare it to the corresponding SDID estimate. As the outcome of interest $Y_{c,t}$, I use the estimated total number of housing units per 1,000 people in a given city and year. Accounting for population in the outcome again mitigates the influence of baseline population differences and growth trends across cities. I choose to begin the time period of interest for this specification in 2010. This time period is selected as a natural normalization point for the housing market following the Great Recession. Additional robustness checks are not so necessary in this specification; the synthetic control estimate is always compared to an SDID estimate with “optimal” time period weights, and the inclusion of covariates is not typically regarded as necessary in either specification.

V. Results & Discussion

V.1 Historical SEPA Results

Table 5 reports the results of synthetic difference-in-differences estimation of the historical effect of passing a SEPA across the several treatment and donor blocks described in Table 3, along with corresponding bootstrapped t-statistics (3,000 replications).¹⁰ No adjustment for multiple hypothesis testing has been made to the t-statistics, and most of the estimates fail to meet conventional benchmarks for statistical significance. Fortunately three important trends hold that make the estimates in Table 5 relevant to the hypothesis that SEPA litigation risk drives a real effect on residential permitting and construction.

¹⁰I plan to run this again with more replications (10,000) but ran out of time to meet the second-year paper deadline.

Table 5: Main Spec SDID Results

	(1) Units/1K	(2) SFH/1K	(3) Mid/1K	(4) Big/1K
<u>SEPA Fixers (CA CT MA MN NY WA)</u>				
Donors: Some EIR	-1.461 [-1.57]	-0.420 [-1.23]	-0.118 [-1.54]	-1.002 [-1.87]
No EIR	-1.612 [-1.93]	-0.511 [-1.22]	-0.214 [-1.77]	-1.264 [-2.28]
All non-SEPA	-1.602 [-1.87]	-0.407 [-1.25]	-0.136 [-1.94]	-1.073 [-1.98]
<u>SEPA Block 1 (CA MN NY WA)</u>				
Donors: Some EIR	-1.064 [-0.79]	-0.407 [-0.83]	-0.098 [-1.02]	-1.163 [-1.46]
No EIR	-1.011 [-0.92]	-0.346 [-0.66]	-0.202 [-1.19]	-1.431 [-1.89]
All non-SEPA	-1.178 [-0.95]	-0.351 [-0.73]	-0.113 [-1.34]	-1.257 [-1.58]
<u>SEPA Block 2 (CT HI MA WI)</u>				
Donors: Some EIR	-2.148 [-1.58]	-0.934 [-1.51]	-0.143 [-1.46]	-0.857 [-1.52]
No EIR	-3.847 [-1.85]	-0.995 [-1.73]	-0.283 [-2.00]	-1.415 [-1.61]
All non-SEPA	-2.557 [-1.75]	-0.775 [-1.52]	-0.164 [-1.44]	-0.922 [-1.64]
<u>SEPA Block 3 (GA IN MD MT NC SD VA)</u>				
Donors: Some EIR	0.765 [1.20]	0.625 [1.98]	0.016 [0.30]	-0.123 [-0.22]
No EIR	0.138 [0.17]	0.621 [1.47]	-0.029 [-0.41]	-0.590 [-0.88]
All non-SEPA	0.507 [0.77]	0.705 [2.22]	0.009 [0.16]	-0.233 [-0.44]

Notes. Units/1K is total housing units permitted per 1,000 population in a given year. SFH refers to single-family homes. Mid refers to permitted units in buildings with 2-4 units. Big refers to permitted units in buildings with 5 or more dwelling units. Fixers are states that have passed laws reducing or eliminating the risk of a lawsuit for residential developers in their state since 2024. Estimates are the average estimated effect of having a SEPA on the given housing measure for each year in the given SEPA block relative to the synthetic trend counterfactuals. Bootstrapped t-statistics (10,000 reps) in brackets.

First, of the 48 estimates presented in Table 5, five are statistically significant at the conventional 95% confidence level with $|t| \geq 1.96$. This suggests a $5/48 = 10.42\%$ Type I error rate if the true effect of SEPAs is null across every estimate reported in the table, or about double the Type I error rate implied by the choice of a 95% confidence level. This suggests that at least some of the estimates are not statistically significant only by random chance; instead of discarding them as random output, we may plausibly interpret at least some of the estimates that align with theory as causal estimates of the effects of SEPAs on permitting.

Second, most of the estimates for the effects of SEPAs on total units in column (1) are close to additive of the estimates for the broken-out unit types in columns (2)-(4). This suggests that the synthetic trends generated to examine the full set of permits for each block were not so different than the synthetic trends generated to examine each of its component parts, which in turn allows for a more “apples-to-apples” comparison of coefficients than would otherwise be possible. This is especially reassuring when presenting the individual unit and time weights for many specifications is prohibitively cumbersome, as is the case here.

Third, the estimates of the effects of SEPAs that use the No EIR donor pool are universally more negative than the estimates using the Some EIR and All non-SEPA donor pools. In fact, the order of the majority of sets of three estimated effects is, in order from positive to negative, Some EIR, All non-SEPA, and No EIR. Table 3 already showed that the average No EIR state trends differently from averages in all other blocks by increasing total and single-family permits issued after 1975, so this effect ordering is not unexpected in those columns. The fact that this persists for housing units permitted in mid-sized developments (2-4 units, column (3)) and big developments (5+ units, column (4)) may suggest reason for cautious interpretation of presented coefficients that include No EIR states. On the other hand, No EIR states comprise the only donor pool in which a lawsuit predicated on a residential developer’s public EIR disclosure is impossible, so these estimates may actually provide the most compelling counterfactual given the binary SEPA-on or SEPA-off assignment of treatment used here.

Even taking these trends under careful consideration, it remains to be seen whether any of the statistically significant coefficients are economically meaningful. First, consider the estimates of -1.264 and -1.073 units permitted in large multifamily residences per 1,000 population per year in SEPA Fixer states. These estimates suggest a decline in the permitting of the exact type of development most likely to attract the ire of local opponents; large apartment complexes and high-rise condo towers are in this category. Relative to the mean reported in Table 3 for fixer states from 1960–1975, these estimates blame SEPAs for a 39%–46% drop in big building permitting rates from pre-SEPA levels. In the same period, permits in these types of buildings made up approximately $2.76/7.41 = 37\%$ of the residential permit supply, so the overall effect of SEPA litigation risk may have been to restrict the total permit supply by between 14%–17%. Given the mild suspicion of the No EIR donor pool explained previously, I present the estimate using All non-SEPA states (-1.073) as my preferred estimate.

The economic implications of this estimate are large. Table 6 presents an important (albeit naïve) extrapolation of this estimate. For example, as show in the table, the preferred estimate suggests that California would have permitted a total of 850,892 more residential units in large multifamily structures between 1970 and 1999 than they actually did. A 2020 Freddie Mac estimate of the housing undersupply in California puts the level of undersupply at 820,000 missing units, while other estimates range between 56,000 and 3 million (Christopher, 2025). The results of this analysis therefore suggest that SEPA litigation risk had a massive effect on residential permitting patterns in California. The table also shows that the other five states in the “fixers” block would have seen a significantly larger housing supply under the same counterfactual circumstances. This counterfactual exercise is optimistic in that it assumes a 100% permit-to-construction rate, but it nevertheless demonstrates that the effects of SEPA litigation risk on residential development may have been quite large.

Next, consider the estimate of -0.283 units permitted in mid-sized multifamily buildings (2-4 units) in SEPA Block 2. Though this estimate is statistically different from 0 at conventional levels, there are practical reasons to doubt it as a true causal effect. For one, middle housing building permits are already unpopular across the country in the studied timeframe, so there is a higher chance than usual that noisy data makes fitting a convincing synthetic trend unreliable. For another, it suggests an over 50% drop in the yearly permit rate from the pre-SEPA period for these types of units. This seems too large to be realistic, especially given the smaller effect already observed in a higher litigation risk block of states for developments

Table 6: Counterfactual Permit Counts

	CA	CT	MA	MN	NY	WA
Years, SEPA $\rightarrow$ 1999	30	27	28	27	25	29
Avg. Units in Big Builds	58,522	3,231	6,178	6,021	9,592	11,422
Avg. Counterfactual Units	86,885	6,660	12,512	10,609	28,811	16,255
Avg. Yearly SEPA Effect	-28,363	-3,429	-6,335	-4,588	-19,220	-4,833
95% Lower Bound	-1,697	-205	-379	-275	-1,150	-289
95% Upper Bound	-66,977	-8,097	-14,959	-10,834	-45,386	-11,413
Total Estimated SEPA Effect	-850,892	-92,579	-177,373	-123,876	-480,491	-140,165

Notes. Big Builds = buildings with 5 or more residential units. Counterfactual units calculated as $((\text{Units per 1K}) - \hat{\tau}_{\text{SDID}}) * (\text{Population in thousands})$. Bounds subtract $\hat{\tau}_{\text{SDID}} \pm 1.96 * \text{SE}_{\tau}$.

more likely to attract lawsuits.

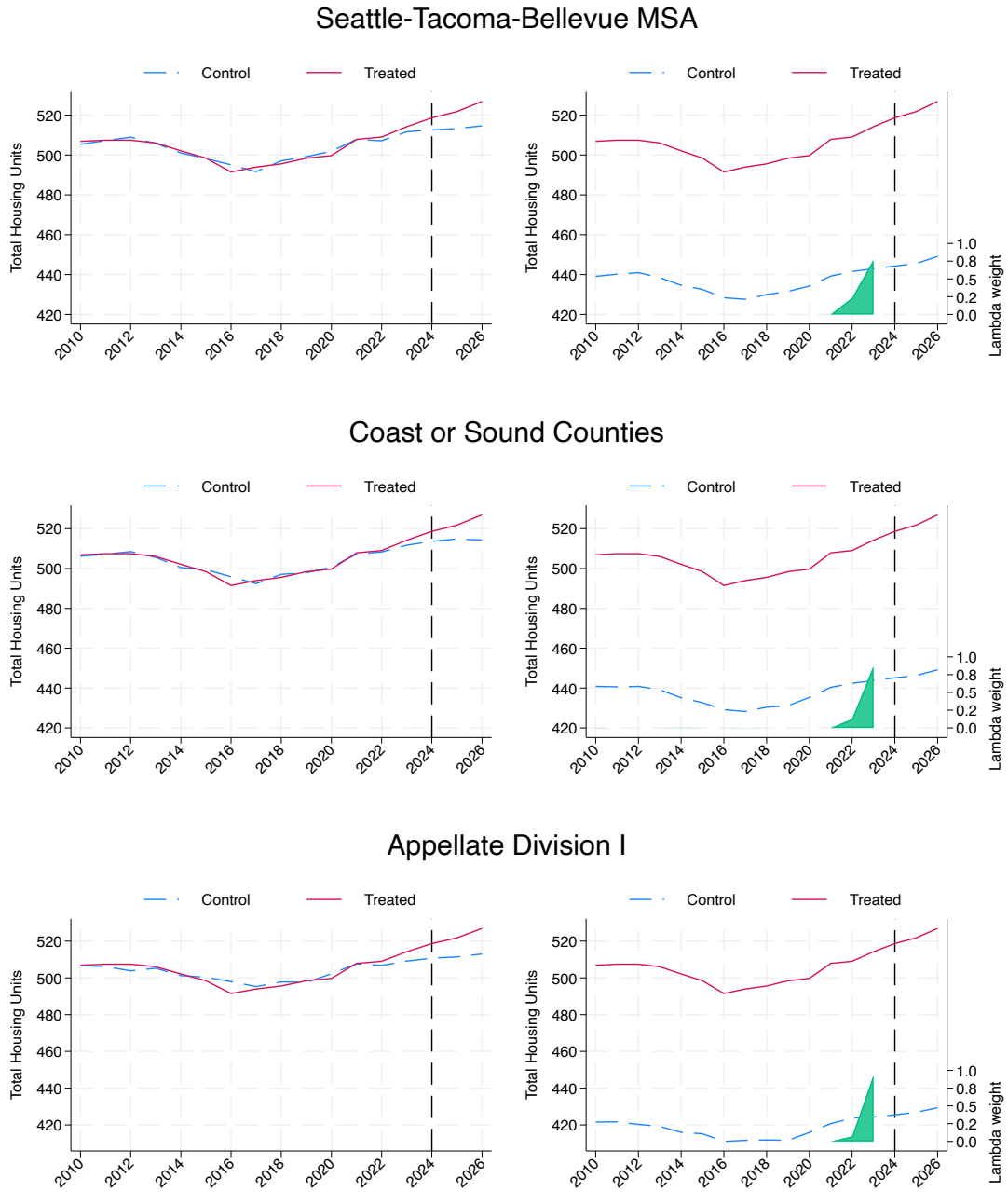
Finally, consider the estimates of +0.625 and +0.705 units permitted in single-family residences per 1,000 population per year in SEPA Block 3. These estimates imply that developers in states with SEPAs that have never sued a residential developer seek to build statistically more single family housing than developers in other states. Given the averages reported in Table 3, these estimates represent about a 10% increase in population-adjusted single family permitting rates. While this would represent a moderate increase, the estimates are in the opposite direction one would expect under the litigation risk theory. At worst, a SEPA that introduces no litigation risk to a residential developer should have no effect on their permitting behavior; the SEPAs of states in Block 3 only require EIR for state actions, not local actions, so most residential development should be unaffected. If the effect was to shift construction from larger developments to single-family homes, perhaps because larger developments are more likely to need state action subject to a SEPA, then we would expect to see a corresponding shift away from multifamily units that is not present in the SEPA Block 3 results in Table 5.

V.2 Seattle Holiday Results

Figure 4 presents the conventional synthetic control and synthetic difference-in-differences figures associated with estimating the effect of the Seattle environmental review holiday on the total estimated housing supply per 1,000 population. Visual inspection of the figures suggests a good fit under both matching methods. Though the figures seem to diverge before treatment, note that the figures mark the first period in which treatment is active, not any pre-period, so apparent divergence “before” treatment may be exaggerated.

Table 7 displays the associated average treatment effect estimates with bootstrapped t-statistics. The treatment effect estimates themselves are uniformly positive and quite large: the median synthetic control estimate of the effect of Seattle’s review holiday is an average increase in 8.957 housing units per 1,000 people in Seattle per year for 2024–2026. Extrapolating to the statistically unlikely counterfactual where this estimate is indeed the true effect suggests that the Seattle review holiday added over 21,800 housing units to the Seattle housing supply in just three years, or about half of the average growth of the housing supply between 2020–2023 and 2024–2026 reported in Table 4. The synthetic difference-in-differences estimates return a more conservative median estimate of 5.669 housing units per 1,000 people, or about 13,800 housing units under the same counterfactual. Though both methods for analyzing the Seattle review holiday suffer from a lack of statistical power, the direction and magnitude of the estimated effects are both reasonable, and they add some evidence that removing litigation risk from the development process can spur construction.

Figure 4: Synthetic Seattles



Notes. Black dotted lines mark the first treated period, not any time before treatment.

Table 7: Seattle Review Holiday Results

	(1) SC	(2) SDID
<i>Seattle-Tacoma-Bellevue MSA</i>		
Holiday Effect, 2024-2026	8.957 [0.22]	5.748 [1.61]
<i>Coast or Sound Counties</i>		
Holiday Effect, 2024-2026	8.190 [0.64]	5.669 [1.25]
<i>Appellate Division I</i>		
Holiday Effect, 2024-2026	10.743 [0.28]	5.658 [1.72]

Notes. SC = synthetic control; SDID = synthetic difference-in-differences. Estimates are the average estimated effect of the holiday on total housing units per 1,000 population present each year in Seattle relative to the counterfactual. Bootstrapped t-statistics (10,000 reps) in brackets.

Section VI. Conclusion

What is the optimal level of public participation in the development process? Apparently, somewhere below the status quo in several states. In the 1970s, many states followed national policy and established state environmental policy acts (SEPA). These laws instituted environmental impact review (EIR) requirements for government projects and mandated public disclosure of potential impacts. In some of those states, courts determined that permitting constitutes a government project, exposing private residential developers to SEPA lawsuits during the permitting stages of their projects. This paper provides evidence that the litigation risk imposed on developers in those states had a substantial deterrent effect on the permitting of large residential projects. A preferred estimate attributes about half of the average decline in permitting housing units of this kind between 1960–1975 and 1976–1999 to SEPA litigation risk.

For those who have argued that their states’ SEPA that open private residential developers to lawsuits go too far, it is no surprise that the most pronounced negative effects of litigation risk occur for very large developments (5+ units) in states that later felt the need to reduce the risk of legal action against residential developers created by their SEPA. Large developments are the most likely to attract attention from local opponents, and the publicity created by their lawsuits can create real political pressure. However, it does not appear to be the case that a few sensational one-off SEPA lawsuits spurred policymakers to unnecessarily weaken iconic environmental protection laws, as is sometimes argued by SEPA proponents. Rather, the results shown in this paper suggest that policymakers in those states have felt real pressure to pass SEPA fixes due to an economically and statistically significant effect of litigation risk that has dissuaded residential development over time.

The results of this paper suggest that residential developers behave like firms in other industries: under greater litigation risk, they act more cautiously. When that risk is removed, like when Seattle was given an environmental review holiday beginning in 2023, preliminary evidence in this paper suggests that builders readjust and build more housing. Policymakers interested in spurring housing development in SEPA states could therefore consider several policy options. They may consider implementing an exemption for residential construction like Seattle’s review holiday or New York’s 2026 carveout for previously disturbed sites. If they value preserving the public reporting requirements that are among the core purposes of the SEPA, they may follow Minnesota’s 2025 decision to reduce the set of people with legal standing for a SEPA lawsuit, or impose short time limits for SEPA challenges as Connecticut did in 2025. In any case, these results strongly suggest that any policy effort that reasonably reduces uncertainty in development cost or timelines could increase developers’ willingness to develop residential housing in SEPA states.

Further research into the effects of SEPAs is warranted. For one, the costs of the reduced housing supply incurred by potential housing market entrants may have been offset by benefits incurred by existing homeowners, perhaps via an increase in asset values. The effect of SEPAs on other kinds of development projects like commercial and manufacturing facilities is outside the scope of this paper but may be well worth investigating. The impact of environmental impact review requirements on actual environmental outcomes is also not well understood. All of these effects represent missing information relevant to both the design of optimal policy in the housing space and in determining the optimal level of public participation in government broadly.

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